



Procedure

TESTING AND VERIFICATION

The Procedure for Testing and Verification is the 11th of 19 procedures under the New Change Management Process. It applies to NNIT employees responsible for testing and verification of system changes prior to implementation to Non-PROD and PROD environment.



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1 Introduction

The Procedure for Testing and Verification is the 11th of 19 procedures under the [New Change Management Process](#). It applies to NNIT employees responsible for test and verification of system changes prior to implementation to Non-PROD and/or PROD environment.

1.1 Acronym

The following acronyms are contained in this document. This is a reference from time to time going through the content.

Term	Definition
PROD	Production
QMIT	Quality Management Information Technology

1.2 Purpose

The purpose of procedure for testing and verification is to ensure the following:

- Changes are tested and verified prior to implementation to Non-PROD environment; and
- Changes are tested and verified prior to implementation to PROD environment

1.3 Scope

The scope of procedure for testing and verification covers input and output to and from the procedure. The details are as follows:

- **Input to the procedure**
 - Test/Verification plan (test cases) for Non-PROD environment (if applicable)
 - Test/Verification plan (test cases) for PROD environment
- **Output from the procedure**
 - Test result that passed verification
 - Test result that did not pass verification

1.4 Limitation

The limitation of the procedure for test and verification applies to systems with non-PROD environment. In such case, the scope of testing and verification is then limited only to systems with PROD environment.

1.5 Roles and responsibilities

The table below details the roles and responsibilities for this procedure.

Role	Responsibilities
Change Tester	• Test the change in system and verify implementation in PROD environment • Test the change in system and verify implementation in Non-PROD environment • Document test result • Add more test cases if needed • Initiate fall-back if needed • Create incident if needed
Test Approver	• Ensure that the change has been tested according to plan • Ensure that the test result is documented and verified • Ensure sufficient testing is conducted
Change Coordinator	Prepare the Change for production implementation with the following focus: • Implementation plan • Verification plan • Fall-back plan

2 Procedure sub activities

Testing of changes to be implemented in a Non-PROD environment is the same with the testing and verification in a PROD environment only. Hence, testing and verification have been contained in the same procedure.

The requirements to Execute Test and Qualification Procedure in NNIT has to be followed when designing the test of the changes in both Non-PROD and PROD environment. The requirements in the procedure covers IT System Development Process and the Infrastructure Project Process.

Refer to [Execute Test and Qualification](#) for the detailed description and apply special attention to the documentation requirements for the different system.

2.5 Procedure steps

The following table is a sequence of action and who is responsible for each step. It depicts a normal flow of the procedure without the existence of exceptional conditions such as addition of more test cases, initiation of fall-back, or creation of incident.

Step	Activity	Responsible	Remarks
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Procedure for Testing and Verification

Step	Activity	Responsible	Remarks
1	Perform test and verification as described in the Task/Verification plan	Change Tester	• Non-PROD and PROD environments • PROD environment only
2	Document test result	Change Tester	Verify if successful or not
3a	Notify Test Approver of successful testing and verification	Change Tester	Update time spent
3b	Receive notification and proceed to next step	Test Approver	Next step: 10. Procedure to Quality Approval
4a	Assign to Change Coordinator if testing and verification is not successful	Change Tester	Update time spent
4b	Acknowledge assignment and proceed to next step	Change Coordinator	Next step: 07. Procedure for build

2.6 Procedure diagram

The following are diagrams that illustrates the procedure steps. Two scenarios are highlighted: one for both non-PROD and PROD environments and another for PROD environment only.

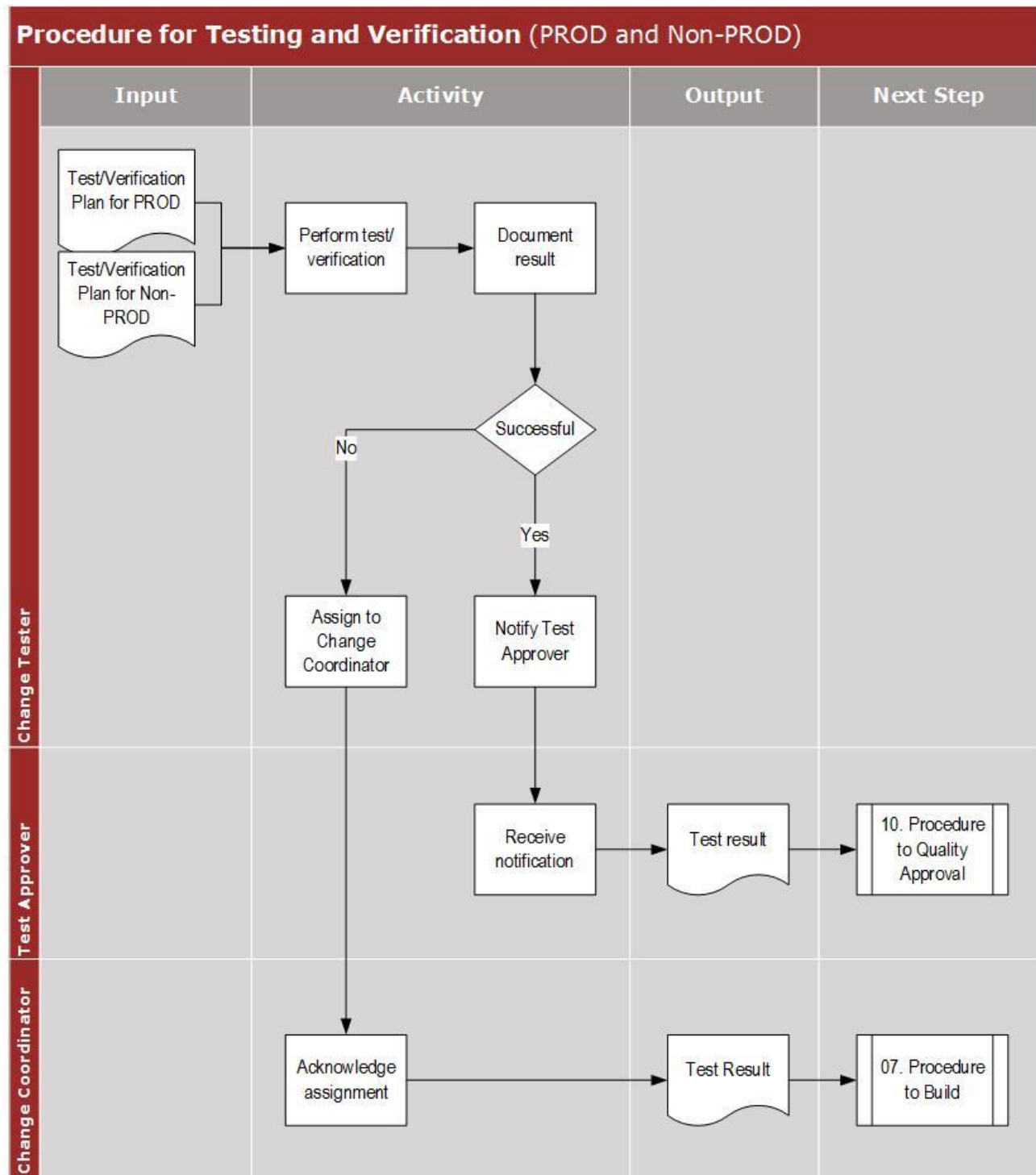


Figure 1. Procedure for Testing and Verification in both PROD and Non-PROD

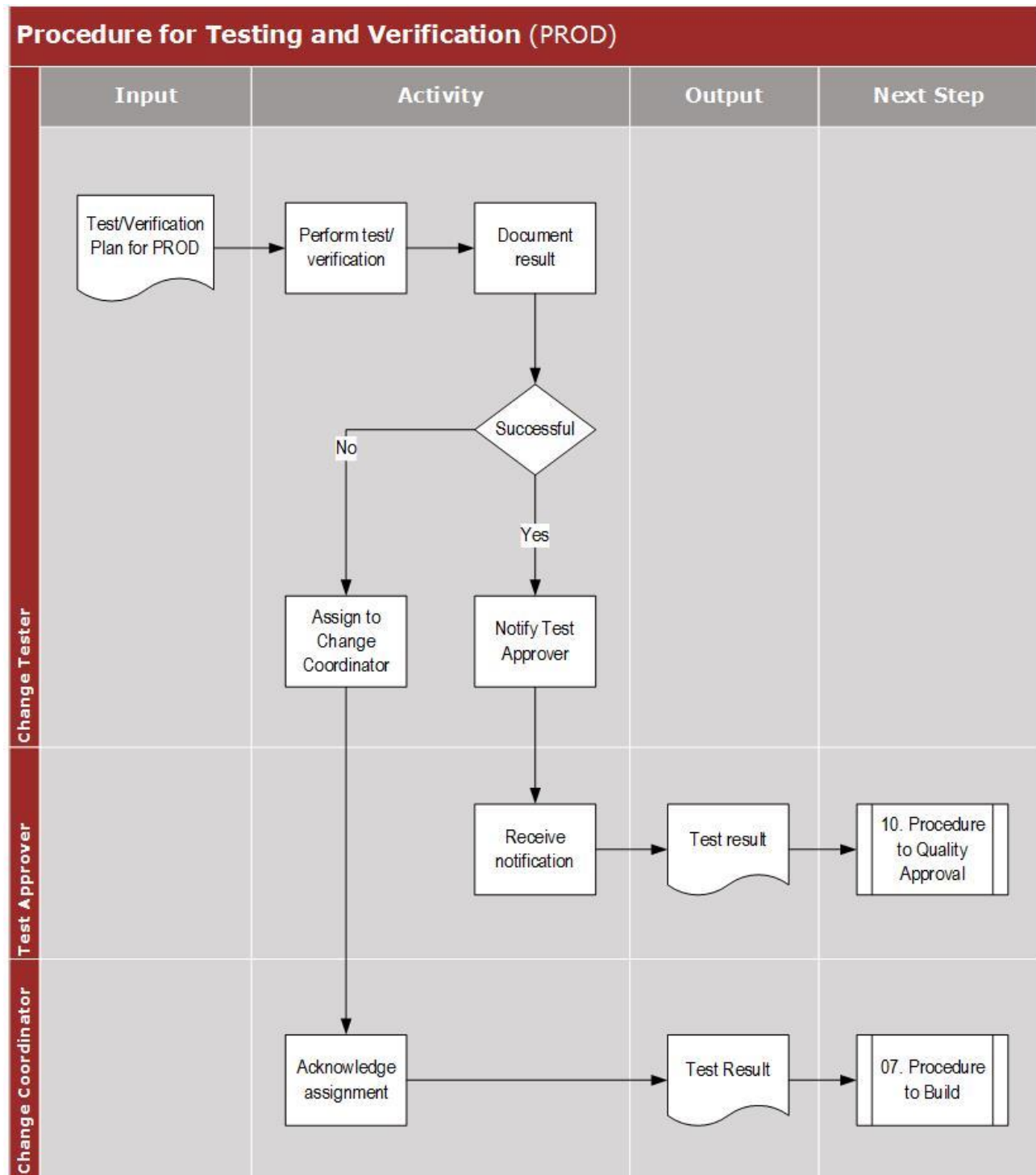


Figure 2. Procedure for Testing and Verification in PROD only